What You Need:

1. **SonarQube Server** – You have already downloaded it. Make sure it is running.
2. **SonarScanner** – A command-line tool that sends analysis reports to SonarQube.
3. **Java & Maven Installed** – Since you have a `pom.xml` file, you should have Maven.

 # Start SonarQube Server

- Open a terminal and navigate to the SonarQube directory.

Start the server:

```
./bin/linux-x86-64/sonar.sh start  # Linux/macOS
```

- Open `http://localhost:9000` in a browser and log in (default username/password: `admin/admin`)

METHOD 1(using sonar scanner) :

Step 1: Download and Install SonarScanner

SonarScanner is needed to analyze your Java project.

Install SonarScanner on RedHat

```
Wget
https://binaries.sonarsource.com/Distribution/sonar-scanner-cli/sonar-scanner-7.0.2.4839-linux.zip
```

1. Extract the downloaded file:

   ```
   # unzip sonar-scanner-7.0.2.4839-linux.zip
   ```

2. Move it to `/opt` for system-wide access:

   ```
   # sudo mv sonar-scanner-7.0.2.4839-linux /opt/sonar-scanner
   ```

3. Add SonarScanner to your `PATH`:

   ```
   # echo 'export PATH=$PATH:/opt/sonar-scanner/bin' >> ~/.bashrc

   # source ~/.bashrc
   ```

4. Verify installation:

   ```
   # sonar-scanner --version
   ```

Step 2: Configure SonarScanner for Your Project

Navigate to your Java project folder and create a `sonar-project.properties` file:

```
cd /path/to/your/java/project   (where  pom file exist or java project root directroy)

vi sonar-project.properties

    sonar.projectKey=my-java-project

    sonar.projectName=My Java Project

    sonar.projectVersion=1.0

    sonar.sources=src

    sonar.language=java

    sonar.java.binaries=target/classes

    sonar.sourceEncoding=UTF-8

    sonar.host.url=http://localhost:9000

    sonar.login=your_sonar_token
```

Step 4: Run Code Analysis

Now, execute the SonarScanner command to analyze your Java code:

```
# mvn compile

# sonar-scanner
```

Step 5: View Results

1. Open **http://localhost:9000** in your browser.
2. Click on your **project name** to view:
    - Code quality reports
    - Bugs & vulnerabilities
    - Code smells & duplications

METHOD 2 (using maven-> mvn sonar:sonar) :

Step 1: Add SonarQube Plugin to `pom.xml`

Open your `pom.xml` file in the project root directory:

```
<build>
    <plugins>
<!--  SonarQube Scanner Plugin  -->
        <plugin>
            <groupId>org.sonarsource.scanner.maven</groupId>
            <artifactId>sonar-maven-plugin</artifactId>
            <version>3.9.1.2184</version>
        </plugin>
    </plugins>
</build>

<reporting>
   <plugins>
     <!--  SonarQube Reports  -->
          <plugin>
          <groupId>org.sonarsource.scanner.maven</groupId>
          <artifactId>sonar-maven-plugin</artifactId>
          <version>3.9.1.2184</version>
      </plugin>
   </plugins>
</reporting>
```

Step 2:Also add this to `pom.xml`

```
<properties>
    <maven.compiler.source>17</maven.compiler.source>
    <maven.compiler.target>17</maven.compiler.target>
    <sonar.host.url>http://192.168.56.15:9000</sonar.host.url>
    <sonar.login>your_sonar_token</sonar.login>
 </properties>
```

Step 3: Run the Maven Command for SonarQube Analysis

```
# mvn clean sonar:sonar
```